Electron Backscatter in Energetic Particle Precipitation: Data Analysis and Simulation

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Key Points:

- The rate of atmospheric backscatter of radiation belt electrons is constrained using data collected from low-Earth orbit.
- An improved Monte Carlo model is used to simulate electron-atmosphere interactions and is validated using the same in-situ data.
- The sensitivity and pitch angle distributions of backscattered populations are characterized using this new Monte Carlo model.

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Abstract

When particles from the radiation belts impinge on the atmosphere, they can be absorbed into the atmosphere or deflected back into the magnetosphere. The deflection of particles back into the magnetosphere is known as backscatter, and it is a key link connecting the atmosphere to the magnetosphere – involving collisions with atmospheric neutrals, magnetic mirroring, the production of secondary emissions, and energy transfer from the particle to the atmosphere. Backscatter is both a feedback mechanism to magnetospheric precipitation drivers and an indirect measure of atmospheric energy absorption, making it an important process to quantify and understand.

In this work, we use data from the Electron Fields and Losses INvestigation (ELFIN) satellites to quantify backscatter rates. We find that backscatter rates vary between $\sim$ 5% during periods of loss cone filling and $\sim$ 90% during periods without loss cone filling. We then compare the ELFIN backscatter data to the results of an updated and improved Monte Carlo-based simulation and find excellent agreement with ELFIN-measured backscatter rates.

Finally, we use our improved Monte Carlo model to characterize the pitch angle and energy dependence of backscatter and the pitch angle distributions of backscattered electrons, finding results consistent with previous modeling efforts.

Plain Language Summary

Near-Earth space is filled with high-energy radiation propagating in a variety of different directions. When this radiation impacts the Earth's atmosphere we can experience a number of adverse effects, including the depletion of the ozone layer and disruptions to telecommunications and power distribution. The Earth's atmosphere and magnetic core help deflect this radiation, but it is not currently known exactly what percentage of incoming radiation breaches these defenses and is deposited in the atmosphere.

In this paper, we use a radiation detector on an Earth-orbiting satellite to determine exactly how much radiation the Earth's atmosphere and magnetic field repels. We find that the amount of radiation repelled depends on the angle at which the radiation approaches the Earth. Only about 5% of radiation that is directed straight at the Earth gets repelled, while up to 90% of incoming radiation at shallower angles is repelled. We then use computer simulations to find that the Earth's atmosphere is responsible for repelling the direct radiation, while the Earth's magnetic field is responsible for repelling the radiation at shallow angles.

Studying how often and by what method spaceborne radiation is repelled helps us prepare for extreme events such as solar storms more effectively by allowing us to forecast how much of the radiation they emit will reach crucial infrastructure systems on the ground – for example, solar storms can alter the angle that radiation approaches the Earth at. The information in this paper allows space weather forecasters to better predict and mitigate the effects of space radiation impacting the Earth.

1 Introduction

Energetic particle precipitation (EPP) is the process where charged particles from the Earth's magnetosphere collide with the atmosphere. In this process, particles either deposit their energy in the atmosphere or are deflected back into the magnetosphere via magnetic mirroring and collisions with atmospheric neutrals. Atmospheric deposition of energetic particles causes cascade effects such as the production of odd hydrogen and odd nitrogen that can in turn reduce ozone concentrations in the stratosphere (Randall et al., 2005; Seppälä et al., 2007; Sinnhuber et al., 2012; Funke et al., 2014; Koskinen & Kilpua, 2022). Additionally, EPP is a central loss process for the radiation belts (Ripoll

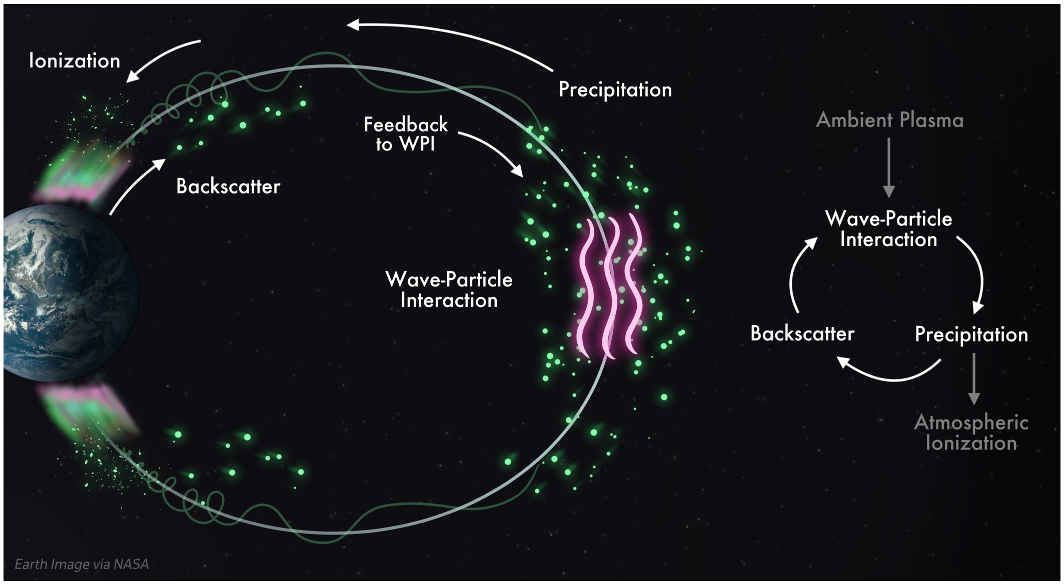

Figure 1. Schematic representation of the interconnected magnetosphere-atmosphere system. Backscatter is a key component of this system, acting as a feedback mechanism that closes the loop between the atmosphere and magnetosphere.

et al., 2020; Millan & Thorne, 2007). The deflection, or scattering, of precipitating particles back into the magnetosphere is known as backscatter. Backscatter is an important process in the magnetosphere, serving both as a feedback mechanism that directly shapes the characteristics of many observed precipitation patterns – such as bouncing packets, diffuse aurora, and lightning-induced electron precipitation (Feinland et al., 2024; Wetzel et al., 2024; Khazanov & Chen, 2021; Cotts et al., 2011; Davidson & Walt, 1977) – and as an indirect measure of atmospheric energy absorption due to EPP. However, until recently, no datasets have been produced with sufficient angular resolution to quantify backscatter on a global scale. This deficiency has been overcome in recent years with the launch of the Electron Losses and Fields INvestigation (ELFIN) mission (Angelopoulos et al., 2020).

In this paper, we quantify the rates of EPP backscatter using data from the ELFIN satellites; compare those results with simulations using an updated and improved Monte Carlo-based EPP model and evaluate the model's performance at predicting backscatter; and use our model to characterize the energy and pitch angle characteristics of backscatter.

2 Background

The existence of backscatter has been known for nearly as long as the radiation belts themselves. In the early 1960s, a sounding rocket experiment measured upgoing auroral fluxes with nearly the same intensity as downgoing fluxes in the lower thermosphere ($\sim$ 100 km) (McDiarmid et al., 1961), sparking interest in the phenomenon. Backscatter was observed on spaceborne platforms shortly after, with backscatter ratios (backscattered flux divided by precipitating flux) observed on the order of 10% (O'Brien, 1962, 1964). Further interest was generated by Cummings et al. (1966) with the discovery that backscatter showed a dependence on precipitating flux, ranging from approximately 20% at high fluxes to nearly 100% at low fluxes. Richards and Peterson (2008) used data from the FAST satellite, finding backscatter ratios for sub-100 keV electrons in the range of approximately 25% at 100 keV up to nearly 100% in the sub-eV range. The high backscatter for lower energy channels indicates a softening of precipitating particle energy spectra.

On the theoretical side, a number of simulation studies have been conducted to better understand EPP backscatter (Maeda, 1965; Cummings et al., 1966; Walt et al., 1968; P. Banks & Nagy, 1970; P. M. Banks et al., 1974; Berger et al., 1974; Mantas & Walker, 1976; Davidson & Walt, 1977; Lejeune, 1979; Cotts et al., 2011; Marshall & Bortnik, 2018; Berland et al., 2023). Techniques for both simulating and reporting backscatter statistics vary from author to author. For example, early models (Maeda, 1965; Berger et al., 1974) did not incorporate the effect of magnetic mirroring. Some studies analyzed backscatter as a function of pitch angle (Berger et al., 1974; Cotts et al., 2011; Marshall & Bortnik, 2018; Berland et al., 2023), but many opted to study the backscatter of predefined pitch angle distributions as a function of energy, thus neglecting a key sensitivity (Maeda, 1965; Cummings et al., 1966; Walt et al., 1968; P. Banks & Nagy, 1970; P. M. Banks et al., 1974; Mantas & Walker, 1976; Solomon, 2001). In addition to differences in simulation procedure, a number of different quantities have been referred to as "backscatter", further complicating comparisons (Mantas & Walker, 1976). A comprehensive simulation of backscatter incorporating a realistic magnetic field without prescribed input pitch angle or energy distributions for electrons would not come until Cotts et al. (2011), followed by Marshall and Bortnik (2018) and Berland et al. (2023).

Despite the inconsistencies in simulation and reporting, some general trends emerge from the literature. First, we note that backscatter ratios inside the trapped region are significantly lower in simulations that do not incorporate magnetic mirroring, indicating that mirroring is a critical process in shaping the bounce loss cone. Second, the energy spectrum of backscattered populations is much softer (skewed toward lower energies) than the incoming spectrum, with backscattered fluxes often exceeding downgoing fluxes in low-energy regimes due to the effect of high-energy particles being backscattered at lower energies. This effect is responsible for the near-100%, and sometimes greater than-100%, backscatter ratios observed in-situ (e.g. McDiarmid et al. (1961); Cummings et al. (1966); Richards and Peterson (2008)). Additionally, we notice a trend of generally higher backscatter ratios being predicted and observed at lower precipitating fluxes. Finally, we note that interaction with the atmosphere serves to diffuse a pitch angle distribution with the intensity of diffusion increasing as the input pitch angle distribution becomes more field-aligned.

Backscatter plays a role in shaping a number of precipitation patterns (Wetzel et al., 2024; Khazanov & Chen, 2021; Cotts et al., 2011; Davidson & Walt, 1977). However, it remains an understudied phenomenon in part due to a lack of data with sufficient resolution to accurately constrain it for radiation belt electrons. Direct measurement of EPP distributions in the loss and anti-loss cone has been limited by instrument field-of-view and pitch angle resolution compared to the loss cone width (e.g. RBSP) or instrument attitude relative to the geomagnetic field (e.g. POES). Ground-based measurements of EPP (e.g. PFISR) cannot determine incoming EPP distributions without the use of inversion methods (Juarez Madera et al., 2023; Sanchez et al., 2022; Turunen et al., 2016; Miyoshi et al., 2015). With modern data from low-Earth orbit, however, we now have the tools to both comprehensively quantify backscatter rates and validate backscatter models.

3 Data Source: ELFIN

The Electron Losses and Fields Investigation (ELFIN) mission was a pair of twin CubeSats that launched in 2018 into a polar low-Earth orbit with $\sim$ 450 km altitude and $\sim$ 93° inclination. The satellites had 16 energy channels approximately logarithmically spaced between 50 keV to 6 MeV. The electron detector had a field of view of

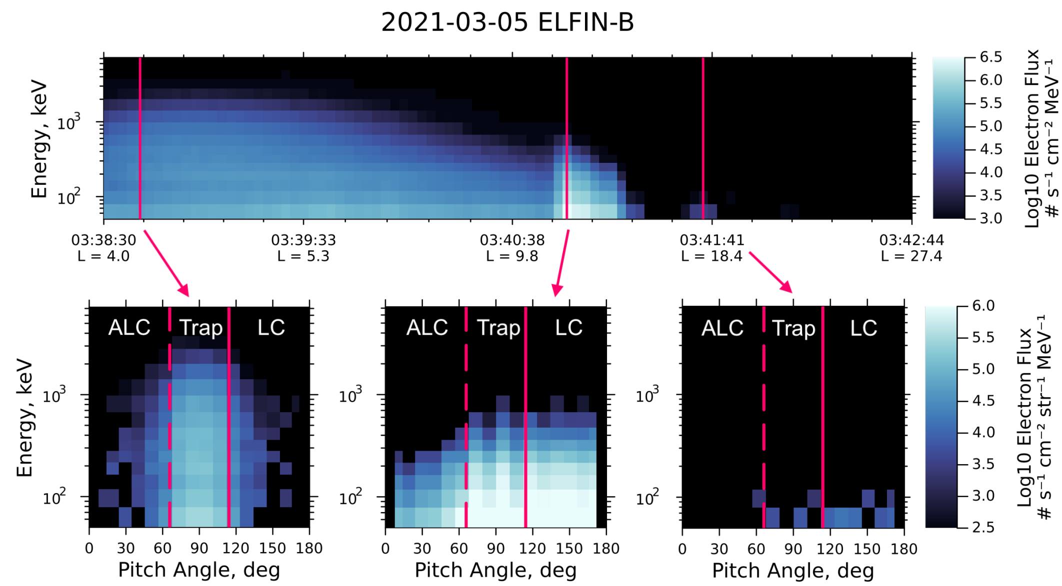

Figure 2. Example data collected from ELFIN. Top: Differential omnidirectional electron number flux as a function of time. Solid red lines indicate the selected timesteps of the bottom panels. Bottom: Differential directional electron number fluxes at selected timesteps. Solid red lines indicate the pitch angle of the 100 km loss cone ("LC") and dashed red lines indicate the pitch angle of the conjugate anti-loss cone ("ALC"), both calculated using IGRF (Angelopoulos et al., 2020). The trapped region is marked as "Trap". This pass was recorded in the southern hemisphere, thus the loss cone angle is greater than 90°. All times are given in UTC.

$\sim 22.5°$ that swept over a full rotation approximately every 2.8 seconds via spacecraft spin. The pitch angle coverage of each spin was dependent on the angle between the spacecraft spin axis and the local magnetic field. The spacecraft spin provided energy- and pitch-angle resolved fluxes during the ELFIN data collection periods discerning the loss cone, trapped region, and anti-loss cone in unprecedented detail (Angelopoulos et al., 2020; Tsai et al., 2025). During ELFIN's science collection periods, the loss cone width, assuming a mirror altitude of 100 km, was approximately 67° in the Northern hemisphere and approximately 113° in the Southern hemisphere. An example of ELFIN data from the Southern hemisphere is shown in Figure 2. The top panel shows omnidirectional differential electron flux recorded by ELFIN over time yielded by multiplying the recorded fluxes at each timestep by ELFIN's solid angle field of view and summing over pitch angle. The bottom panels show the flux measured at three selected timesteps as a function of energy and pitch angle without any integration applied. The loss cone angle is marked with a solid line and the label "LC". The anti-loss cone angle is marked a dashed line and the label "ALC". The trapped region lies between these two lines and is labelled "Trap". The first panel shows ambient fluxes in the outer radiation belt; the second panel, measured in the current sheet, shows an example of pitch angle scattering due to field line curvature into the loss cone with significant backscatter; and the third panel shows a quiet distribution in the polar cap with very little flux.

3.1 Data Selection

For our analysis, all ELFIN data were divided into segments lasting 3 spacecraft spin periods ($\sim$ 8 seconds; $\sim$ 24 km along-track) each. This spin-averaging was done to smooth the data, as ELFIN has occasional artifacts that can be alleviated via integration over multiple spins. Each 3-spin segment was then either discarded or retained for analysis based on its pitch angle coverage. A data segment was kept if it had continuous pitch angle coverage between 5° and 175°, determined using the ELFIN Energetic Particle Detector (EPD)'s nominal field of view of 22.5° (Angelopoulos et al., 2020). This selection criteria ensures that no more than 10° of the pitch angle distribution ($\approx$ 0.05 str) are unrecorded. If any portion of the data segment fell within a period of unreliable data, as provided on the ELFIN data website, the segment was discarded. A known type of corrupted data where all pitch angle look directions measured a uniform high flux was also removed by discarding any data segment with loss cone flux divided by trapped flux ($J_{\text{precip}}/J_{\text{trap}}$) greater than $10^{-0.5}$ and number backscatter ratio (see Section 4 for backscatter ratio calculation) greater than 90%. Noise was then removed within each remaining data segment by zeroing out any data points with relative error greater than 50%, where relative error was computed by the ELFIN team as Poisson noise using Equation 1 (Tsai, 2025). In Equation 1, $\frac{\delta q}{q}$ is the relative error in a given measurement, and N_{counts} is the total number of counts the electron detector recorded during the measurement. After applying these conditions to the lifetime of both ELFIN satellites, we were left with 231 hours ($255,447$ spacecraft spins) of data collected over 2 years. This dataset will be further downselected after the quantification of backscatter in Section 4 to only retain the highest quality segments.

$$\frac{\delta q}{q} = \frac{1}{\sqrt{N_{\text{counts}}}} \tag{1}$$

4 ELFIN Backscatter Rates

In this work, we define the number backscatter ratio r_N as the number of particles backscattered divided by the number of precipitating particles. Similarly, we define the energy backscatter ratio r_E as the amount of backscattered energy divided by the amount of precipitating energy (Equation 2). In ELFIN data, we define input fluxes as any fluxes within the 100 km bounce loss cone and backscattered fluxes as any fluxes within the conjugate anti-loss cone.

$$r_N = \frac{\#_{\text{ALC}}}{\#_{\text{LC}}} \qquad\qquad r_E = \frac{E_{\text{ALC}}}{E_{\text{LC}}} \tag{2}$$

Backscatter rates were derived from the ELFIN data segments via the following procedure. Each data segment was divided into loss cone and anti-loss cone regions based on the central pitch angle of ELFIN's look directions during the segment. Since ELFIN's particle detector field of view (FOV) is $\sim 22.5°$ wide (Angelopoulos et al., 2020), it was determined that a look direction's boresight angle must be more than 11.25° (one half of the instrument FOV) away from the loss or anti-loss cone edge in order to ensure no particles from the trapped region were counted. The loss and anti-loss cone angles were calculated as the pitch angles having 100 km mirror altitudes using IGRF (Angelopoulos et al., 2020; Tsai, 2025); this angle varies geographically by a few degrees over the L ranges probed by ELFIN. Each look direction satisfying these constraints for the loss cone and anti-loss cone was then integrated over energy, pitch angle, area, and time to retrieve the total electron counts and total energy recorded in the loss cone and anti loss cone. The backscatter ratios r_N and r_E were calculated per Equation 2.

After the backscatter ratio was calculated for a given data segment, the error in that value was found by propagating the relative error in the flux measurement at each time, energy, and pitch angle (Equation 1) through the mathematical steps taken to calculate the backscatter ratios. Any value of r_N or r_E with absolute uncertainty greater than $\sigma \approx 0.025$ or fewer than 10 electrons recorded in the loss cone was discarded. The

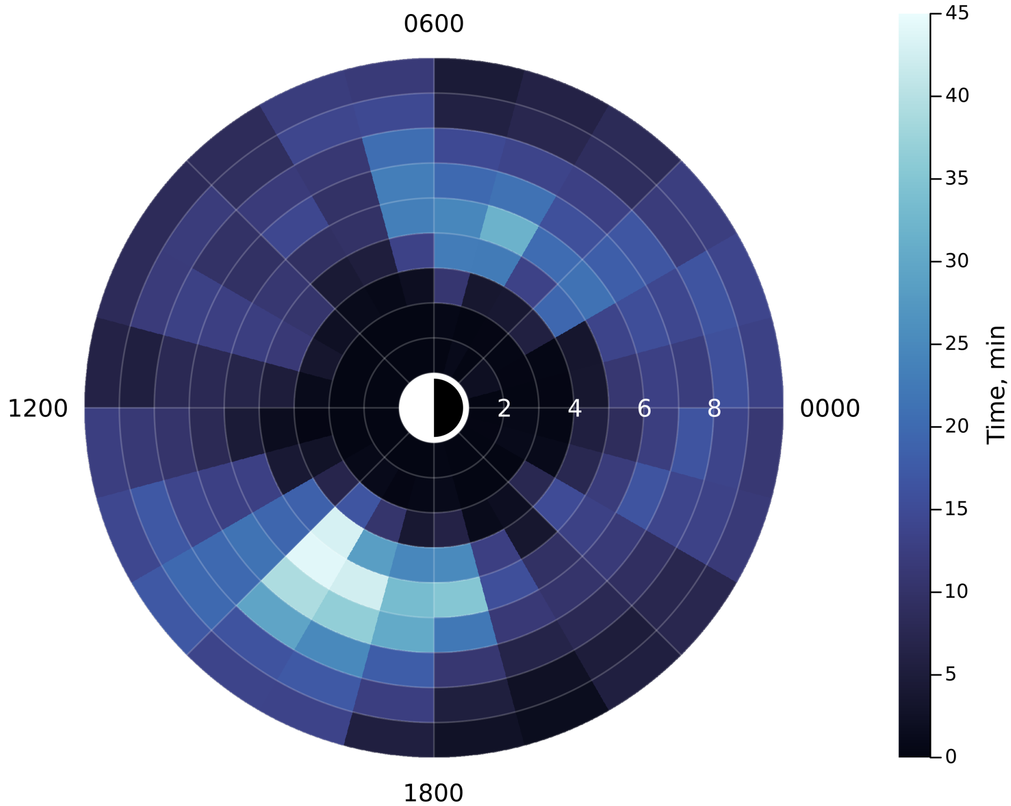

Figure 3. L-MLT coverage of the ELFIN dataset from which we will calculate backscatter ratios. This dataset encompasses 63 hours ($69,510$ spacecraft spins) of ELFIN data collected over 2 years. L and MLT are calculated using the T87 magnetic field model. Some data was collected at $L > 10$ or on open field lines, accounting for more data not shown here.

discarded data is not necessarily unreliable – we are simply being very strict in our selection criteria to ensure the backscatter ratios we analyze are the highest-quality data ELFIN has to offer. This selection criteria resulted in a set of backscatter ratios derived from 63 hours ($69,510$ spacecraft spins) of data collected over 2 years. The coverage of this data in L-shell and magnetic local time (MLT) calculated using the T87LONG magnetic field model (Tsyganenko, 1987) is shown in Figure 3.

The backscatter ratios calculated from this dataset are shown in Figure 4. The backscatter distribution demonstrates three notable features, from left to right in the figure: i) low backscatter ratios at low loss cone fluxes; ii) backscatter rates spanning low and high rates at medium fluxes; and iii) low backscatter ratios at high fluxes.

First we consider the low backscatter ratios at low loss cone fluxes ($\lesssim 10^{2.5}$ electrons-cm^{-2}-s^{-1}). This is not a physical phenomenon, but rather a representation of our error propagation scheme and ELFIN's particle detector sensitivity. Since we are utilizing Poisson noise statistics to determine the uncertainty in the ELFIN measurements (Equation 1), lower fluxes have higher uncertainty. In this region, the fluxes in the loss cone just barely clear our uncertainty threshold to be included in the analysis. Even a small reduction in these fluxes due to interaction with the atmosphere is sufficient to push backscattered fluxes below our uncertainty threshold, whereupon we zero out the backscatter readings for these inputs. In this manner, backscattered particles are missed by falling be-

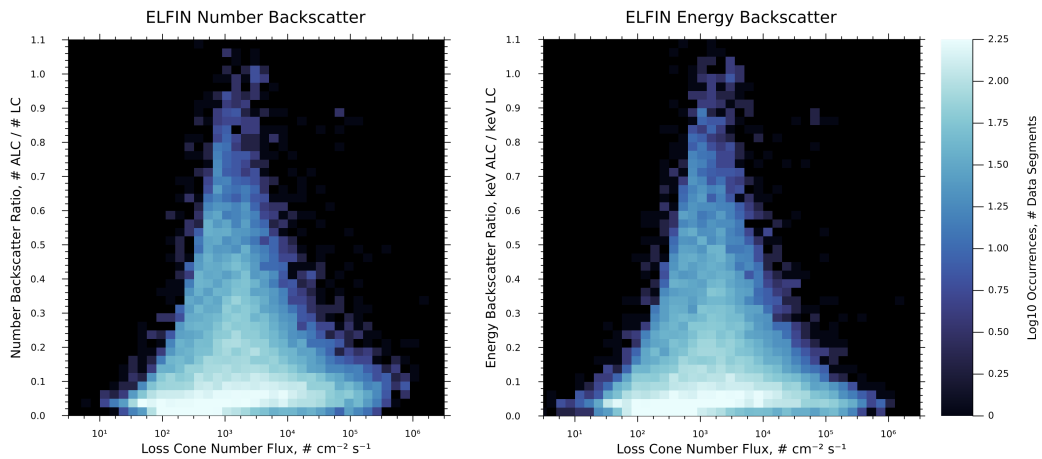

Figure 4. Backscatter ratios derived from 63 hours (69,510 spacecraft spins) of ELFIN data. Backscatter ratio is defined as the total electron number or energy measured in the anti-loss cone divided by the total number or energy measured in the loss cone. Any measurement of backscatter ratio with absolute uncertainty greater than $\sigma \approx 0.025$ was discarded.

low ELFIN's noise floor, creating backscatter ratios with artificially low values for these fluxes.

At high fluxes ($\gtrsim 10^{4.5}$ # cm^{-2} s^{-1}), we see backscatter ratios in the range of approximately $r_N \in [5\%, 20\%]$ and $r_E \in [0\%, 15\%]$. This likely indicates a low rate of backscatter deep in the loss cone. This is because high loss cone fluxes indicate filling via pitch angle scattering processes, which isotropizes the pitch angle distribution and fills the loss cone. The magnetic mirror force is less effective for these more field-aligned particles, sending them deeper into the atmosphere, causing them to be backscattered at lower rates. At this flux level in the loss cone, this pitch angle-scattered population greatly exceeds any ambient populations. Thus, the backscatter ratio for these pitch angle-scattered particles dominates, creating the lower backscatter rates we observe at high loss cone fluxes.

At medium fluxes (between approximately $10^{2.5}$ # cm^{-2} s^{-1} and $10^{4.5}$ # cm^{-2} s^{-1}), we see a spreading of the distribution with backscatter ratios of approximately $r_N, r_E \in [5\%, 90\%]$. We believe the lower backscatter ratios ($\lesssim 20\%$) in this range are due to particles scattered deep into the loss cone, as described previously. We believe the higher backscatter ratios ($\gtrsim 50\%$) indicates much higher backscatter rates near the loss cone edge. When this population is isolated – i.e. when there is weaker pitch-angle scattering – we see much higher backscatter ratios, since any particles deep in the loss cone are lost quickly without a refilling process. This leaves behind only particles near the loss cone edge, indicating that backscatter rates near the loss cone edge can be much higher than the rates at more field-aligned angles. We validate this conjecture in Section 7. Finally, we believe the intermediate range between the high and low backscatter values represent various strengths of pitch angle scattering causing a weighted mix between the high ambient backscatter rates and the lower pitch angle scattering-induced backscatter rates.

If our interpretation of this pattern is correct, we should observe a decreasing backscatter ratio with increased filling of the loss cone and high backscatter rates observed at times with less filling of the loss cone. If we use the ratio of precipitating flux over trapped flux as a measure of pitch angle distribution isotropization, and thereby loss cone filling (e.g. Capannolo et al. (2023)), we indeed observe this pattern (Figure 5, left). Recalling that

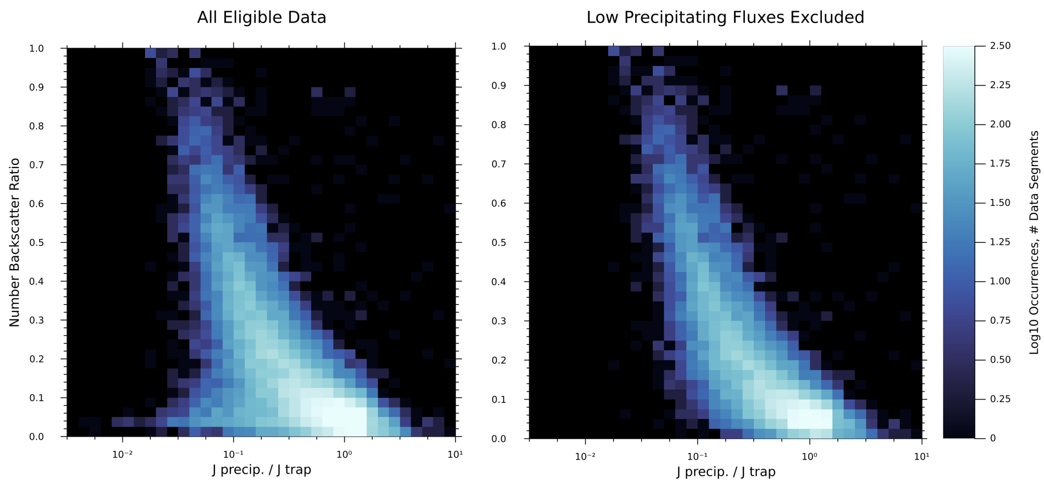

Figure 5. Number backscatter ratio vs. loss cone filling strength derived from 63 hours (69,510 spacecraft spins) of ELFIN data. Any measurement of backscatter ratio with absolute uncertainty greater than $\sigma \approx 0.025$ was discarded. Left: Results using data as previously described. Right: Measurements with precipitating flux less than $10^{2.75}$ electrons-cm^{-2}-s^{-1} removed to avoid artificially lowered backscatter ratios due to backscattered fluxes falling below the instrument noise floor. Energy backscatter ratio has a similar pattern to number backscatter ratio and is not shown for brevity.

low loss cone fluxes cause artificially low backscatter ratios, we remove measurements with loss cone fluxes below $10^{2.75}$ electrons-cm^{-2}-s^{-1} and find a clear pattern confirming our interpretation (Figure 5, right). A numerical fit of the curve in Figure 5 can be found in the Supporting Information.

Additionally, visual inspection of individual data segments at various backscatter levels support the conclusion that high backscatter ratios are present during times of ambient precipitation (i.e. no loss cone filling), low backscatter ratios are seen during times where the loss cone is filled, and intermediate backscatter ratios occur during the in-between cases. Example ELFIN data for each of these situations is shown in Figure 6. The left panel shows the case of ambient populations with an unfilled loss cone, where the majority of precipitating flux resides just inside the loss cone edge. This case has a high backscatter ratio of $r_N = 0.72$, indicating the high backscatter ratios in the region near the loss cone edge. The center panel shows the case of strong loss cone filling. Here, the fluxes reside significantly deeper into the loss cone, with significant fluxes reaching field-aligned angles. This case shows very little backscatter with $r_N = 0.1$, indicating that the backscatter rates deep in the loss cone are much lower than pitch angles near the loss cone. Finally, the right panel shows a data segment with an intermediate backscatter ratio of $r_N = 0.31$, showing a blend of the characteristics of the left and center panels. Some loss cone filling can be observed, but it is not as extreme as the center panel. Thus, we conclude that the dependence of backscatter ratio on loss cone filling indicates the underlying sensitivity of backscatter to pitch angle, with pitch angles near the loss cone edge having high backscatter ratios, transitioning to near-zero backscatter ratios at field-aligned angles.

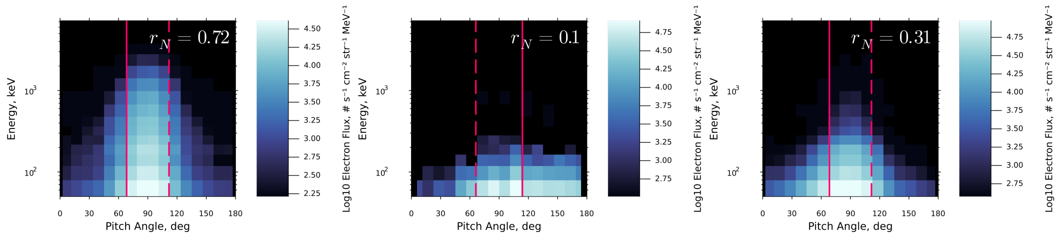

Figure 6. ELFIN-measured fluxes for data segments at three different levels of backscatter. Left: A data segment with little filling of the loss cone and high backscatter ratio. Center: A data segment with strong loss cone filling and low backscatter ratio. Right: A data segment with partial loss cone filling and an intermediate backscatter ratio. Solid lines indicate the pitch angle of the 100 km loss cone, and dashed lines indicate the conjugate anti-loss cone pitch angle. The center panel was recorded in the Southern hemisphere, thus $\alpha_{LC} > \alpha_{ALC}$.

Parameter	**Berland 2023**	**This Work**
Physics List	QBBC	QBBC
Statistical Biasing	100x Bremsstrahlung Splitting	None
Magnetic Field	Uniform	Centered Dipole
Magnetic Latitude	65.77°	65.77°
Electron Injection Altitude	300 km	449.5 km
Electron Collection Altitude	500 km	450.5 km
Minimum Measurable Electron Energy	0.1 keV	0.01 keV
Atmosphere Model	NRLMSISE00	NRLMSISE00
Atmosphere Resolution	1 km	1 km
Atmosphere Model Time	2018-01-01 03:00:00 UTC	2018-01-01 03:00:00 UTC
Atmosphere Model Location (lat, lon)	$(65.0°, -148.0°)$	$(65.0°, -148.0°)$

Table 1. Description of the parameters used in our Monte Carlo simulation, along with the parameters from Berland et al. (2023) upon which our simulation is based.

5 Precipitation Model: Geant4

With backscatter rates constrained by ELFIN data, we now have the ability to directly evaluate the performance of precipitation models. Our approach to precipitation modelling is based on the work of Berland et al. (2023); our model began as a fork from their simulation code. The precipitation model used in this work utilizes Geant4, a kinetic particle simulation code developed by CERN (Agostinelli et al., 2003). Geant4 simulates the interactions of particles with any user-specified material. Here, we utilize Geant4's built-in QBBC physics list, which is the recommended physics list for space radiation applications (Ivantchenko et al., 2012). QBBC includes the effects of bremsstrahlung photon production, an important process at radiation belt energies. For a description of the exact processes and cross sections included in the Geant4 QBBC list, see Section 3 of Berland et al. (2023). No biasing methods such as bremsstrahlung splitting are used in our simulation in order to conserve energy. The parameters used in our model are compared with those used in Berland et al. (2023) in Table 1.

Following the procedure of Berland et al. (2023), we use this model to inject a series of mono-energetic, mono-pitch angle beams consisting of $100,000$ electrons each into the atmosphere from above. These beams were simulated every 5 degrees in pitch angle from 0° to 90°, with the exception of the range from 60° to 70°, where beams were simulated every 1°. This higher resolution accounts for the high sensitivity of backscatter in the pitch angle range surrounding the 100 km loss cone angle ($\approx 67°$ at our injection altitude). The beams were simulated at the mean energies of each of ELFIN's energy channels, from 63 keV to 6500 keV, spaced nearly uniformly in logarithmic space. For each beam, the atmospheric energy deposition and backscattered electron distribution was recorded. A particle was defined as backscattered when it reached the collection altitude with a vertical component of velocity pointed away from the Earth's surface. All backscattered particles were terminated after being recorded. This procedure resulted in a series of tables tabulating the backscattered energies, pitch angles, and momenta for a variety of input conditions, forming a set of backscatter response functions that can be combined in a weighted sum scheme to approximate the backscatter of arbitrary input distributions.

6 Model Validation

In order to compare ELFIN data to our Geant4 modelling, we require a procedure to fit the fluxes recorded by ELFIN to the set of lookup tables generated with our Geant4 model. We first integrated each data segment described in Section 3.1 over time, geometric factor, and energy to produce an electron distribution in energy/pitch angle space with units of electron counts. For distributions measured in the Southern hemisphere, the pitch angle look directions were mirrored about 90° to match Geant4 lookup tables, which were simulated in the Northern hemisphere. At this point, all bins with upgoing pitch angles are discarded. Then, we perform a scaling procedure to account for the fact that ELFIN only measures part of the azimuthal distribution of downgoing electrons. To do this, we assume a uniform distribution of electron flux across all azimuthal angles. We then determine what fraction of the azimuthal distribution ELFIN measures compared to a ring of solid angle between the minimum and maximum pitch angles captured by ELFIN's field of view. At field-aligned pitch angles (i.e. 0° and 180°), this fraction is 1, as all azimuthal angles are measured by ELFIN at those locations. The scaling factor that converts ELFIN-measured counts to the total counts across a uniform azimuthal distribution for a given look direction is called C and is defined by Equations 3 through 6. Figure 7 shows a schematic diagram of the difference between the solid angle measured by ELFIN and the solid angle the simulation propagates.

$$C(\alpha_{\mathrm{bore}}) \equiv \frac{\Omega_{\mathrm{sim}}}{\Omega_{\mathrm{ELFIN}}} = \frac{2\pi\left(\cos\left(\alpha_1\right) - \cos\left(\alpha_2\right)\right)}{2\pi\left(1 - \cos\left(\frac{\theta_{\mathrm{EPD}}}{2}\right)\right)} \tag{3}$$

$$\alpha_1 = \mathrm{clamp}\left(\alpha_{\mathrm{bore}} - \frac{\theta_{\mathrm{EPD}}}{2}, [0, 180]\right) \tag{4}$$

$$\alpha_2 = \mathrm{clamp}\left(\alpha_{\mathrm{bore}} + \frac{\theta_{\mathrm{EPD}}}{2}, [0, 180]\right) \tag{5}$$

$$\mathrm{clamp}(x, [a, b]) = \begin{cases} a & x \leq a \\ x & a < x < b \\ b & x \leq b \end{cases} \tag{6}$$

To simulate the full downgoing electron distribution measured by ELFIN across all azimuthal angles, every downgoing pitch angle and energy bin in an ELFIN data segment locates the nearest pitch angle and energy for which a precomputed simulation beam exists, then adds a weight to that beam equal to the number of counts recorded in that bin multiplied by the value of C at that bin's pitch angle. We repeat this process for every downgoing bin in a half spin of the satellite – weighting the full spin results in a doublecounting of the downgoing distribution, as each half of the spin measures nearly the same pitch angles. After iteration over the half-spin downgoing ELFIN bins, we have a weighted set of beams for which we look up the backscattered electron momenta. To cal-

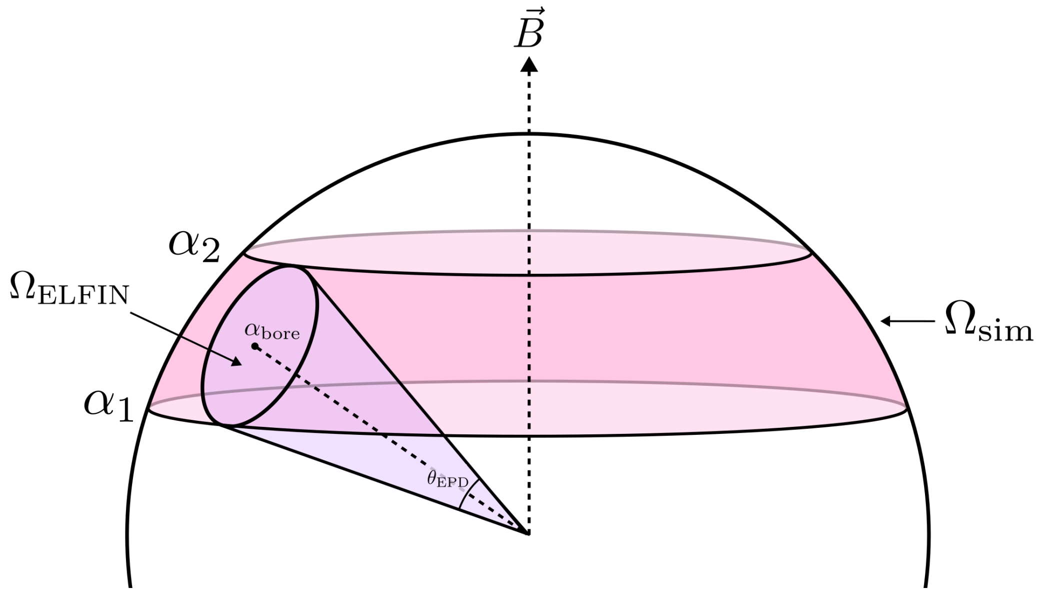

Figure 7. Illustration of the difference in solid angle observed by the satellite (Ω_{ELFIN}) and our simulation (Ω_{sim}) between two pitch angles α_1 and α_2. The simulation propagates all backscatter in a ring between these two pitch angles, while ELFIN only observes a proportional fraction of this ring. Diagram is not to scale.

culate the simulated backscatter rate, we construct a set of virtual ELFIN detectors with azimuthal angles equal to zero and pitch angles equal to the boresight angles of the anti-loss cone look directions in the ELFIN data. Each virtual detector finds the angle between all backscattered electrons and its boresight direction. If this angle is less than or equal to half the ELFIN electron detector field of view (11.25°), the weighted electron is counted as a detection. Backscattered electron detections are also classified by energy into ELFIN's 16 energy channels. The result of this virtual detection process is a simulated backscattered electron distribution on the same grid as the ELFIN data in energy/pitch angle space for the anti-loss cone. Then, similar to our process when cleaning the ELFIN data, any simulation energy/pitch angle bins with relative error greater than 50% as defined in Equation 1 are set to zero for parity with the calculation of backscatter on ELFIN. The simulated counts and simulated energy measured in the anti-loss cone are then summed and stored as a single value. The process of simulating ELFIN backscatter is illustrated in Figure 8.

After simulating the backscatter for all data segments that were analyzed in Section 4, the total anti-loss cone electron flux predicted by our simulation vs. the anti-loss cone count measured by ELFIN were plotted against each other in Figure 9. We can see that the simulation has overall excellent agreement with the backscatter rate data, with a tendency to overestimate backscatter at very low fluxes. This overestimation at low fluxes is not altogether unexpected, as low fluxes cause ELFIN to measure artificially low backscatter ratios due to its noise floor as discussed in Section 4. While we attempted to account for this effect by including a noise floor cutoff in our simulated data, this did not perfectly replicate ELFIN's noise floor. This is likely due to the fact that precisely replicating the number of counts in a given pitch angle and energy bin is a much more demanding task than replicating the counts in the entire loss cone over all energy channels, and the ability to replicate a backscatter distribution on a per-bin basis is limited by the pitch angle resolution that ELFIN measured the downgoing distribution with. Thus

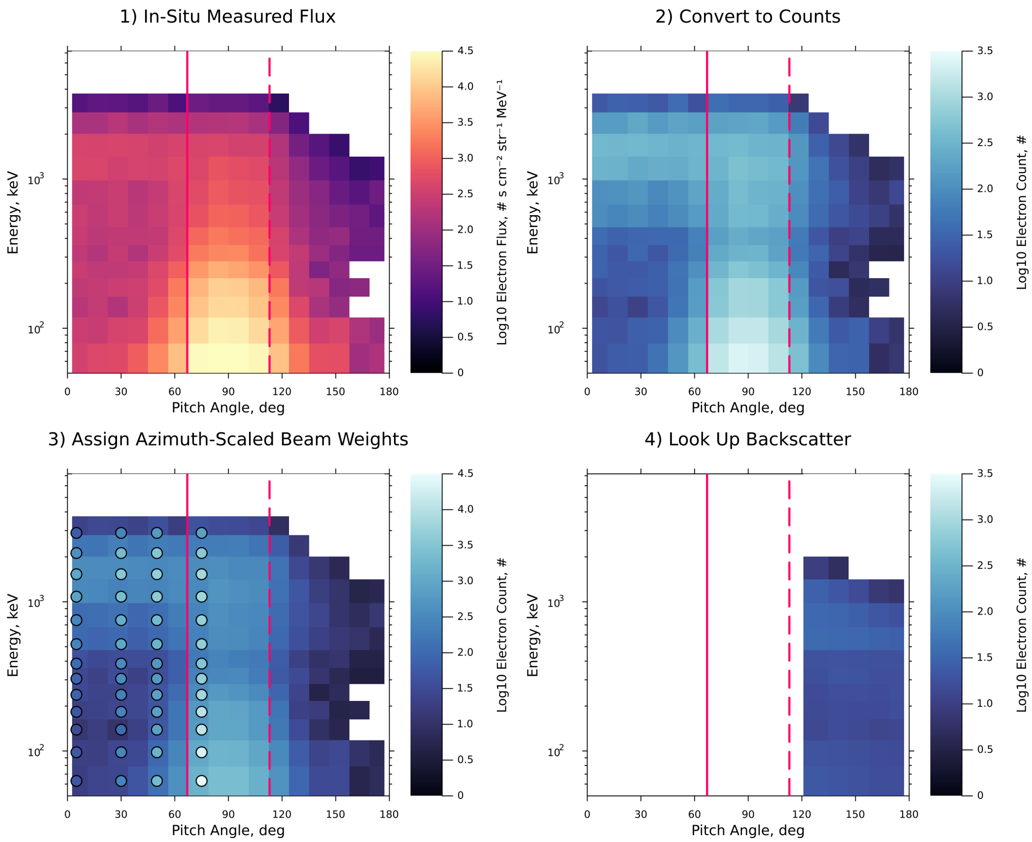

Figure 8. Illustration of the procedure used to predict backscatter based on ELFIN data. Solid lines mark the loss cone pitch angle and dashed lines mark the anti-loss cone pitch angle. 1) ELFIN-recorded fluxes. 2) Each recording bin is multiplied by energy width, instrument geometric factor, and dwell time to produce electron count. 3) Each data bin in the loss cone is scaled by azimuthal field of view and assigns its number of counts as a weight to the nearest energy and pitch angle for which a backscatter lookup table exists. Circular markers indicate the location of these lookup tables, and their color indicates their weight in electron counts. Only every other pitch angle bin is simulated to avoid doublecounting the downgoing distribution. 4) Backscattered electron detections are determined from the simulation-derived weighted lookup tables via a virtual detector for look directions in the anti-loss cone.

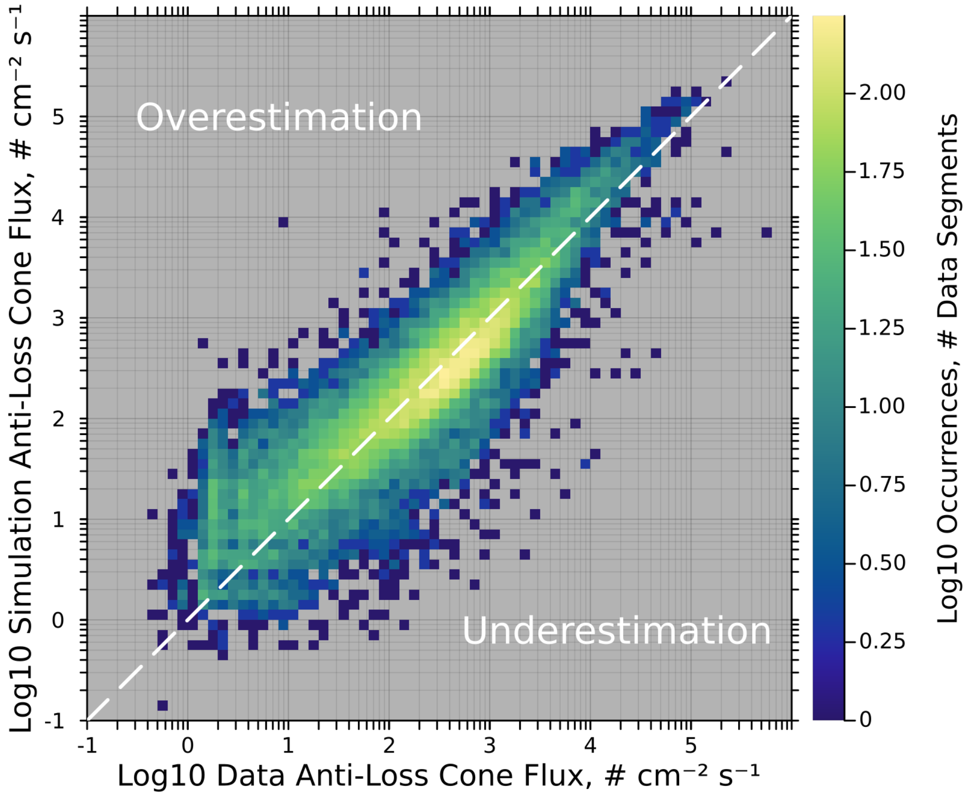

Figure 9. Anti-loss cone electron flux predicted by lookup tables derived from our Monte Carlo simulation vs. ELFIN-measured anti-loss cone electron counts for all ELFIN data segments that were simulated in Section 4. The dashed white line indicates where predicted backscatter was equal to ELFIN-measured backscatter. This is based on 63 hours ($69,510$ spacecraft spins) of data.

we conclude that the overestimation at low fluxes is unlikely to indicate a fundamental issue with our simulation, as the simulation performs excellently at all other flux levels.

7 Simulated Backscatter Characteristics

Using the backscatter lookup tables derived from our Monte Carlo simulation, we can analyze the characteristics of backscatter. In this study, following on the work of Marshall and Bortnik (2018), we investigate the pitch angle and energy dependence of backscatter along with the pitch angle distribution of backscattered electrons. Figure 10 illustrates our predicted backscatter ratios as a function of energy and pitch angle. First, we note the extremely sharp increase in backscatter ratio near the 100 km loss cone angle of 67°. When magnetic mirroring was removed from the simulation, this steep gradient disappeared entirely, confirming that the loss cone boundary observed in this simulation is primarily due to the effects of magnetic mirroring. Secondly, we note that the number backscatter is strictly greater than the energy backscatter. This is due to the fact that for a monoenergetic electron beam, electrons can lose energy to the atmosphere but still be backscattered, causing a reduction in energy backscattered without impacting the total number backscattered. On the other hand, it is not possible for an electron to gain energy in the atmosphere in this simulation, meaning that it is not possible for the energy backscatter ratio to exceed the number backscatter ratio for a monoenergetic in-

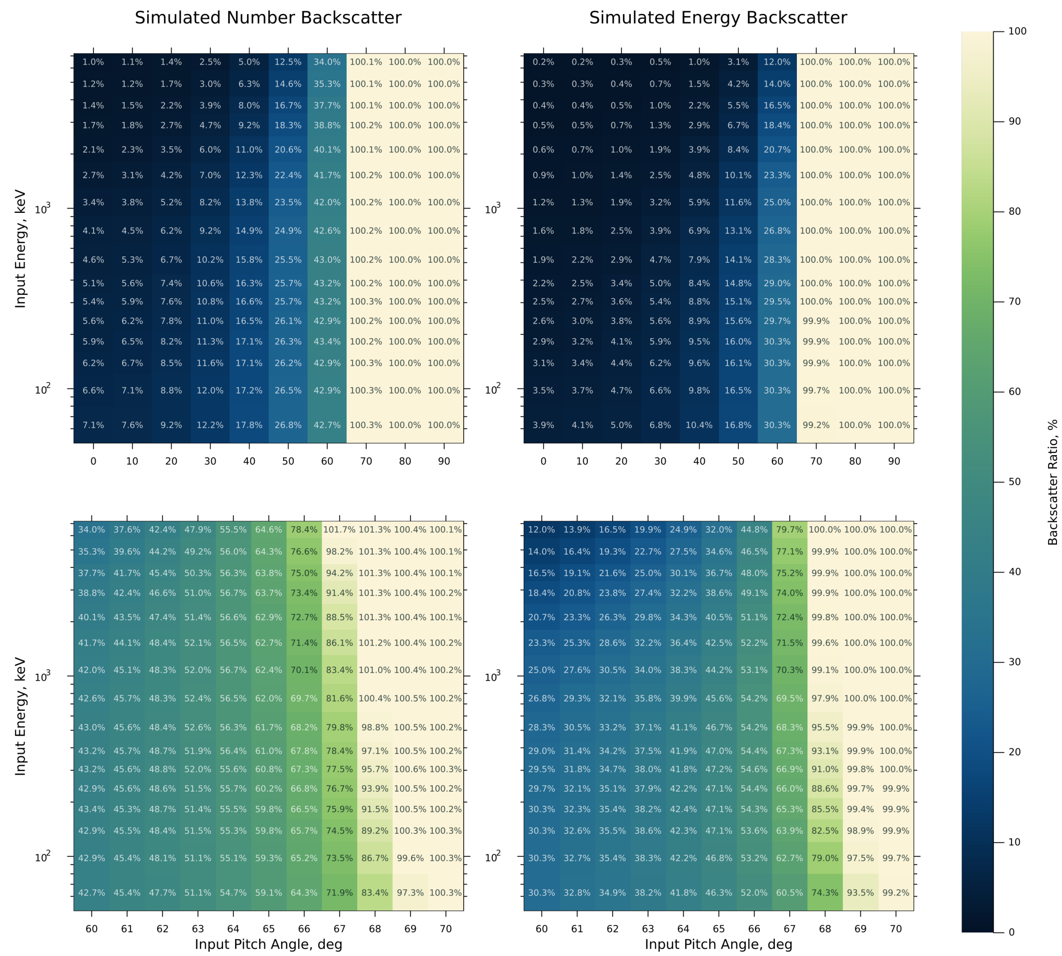

Figure 10. Backscatter ratios for various simulated beams consisting of 100,000 monoenergetic, mono-pitch angle electrons injected into our Geant4 simulation as a function of energy and pitch angle. Electrons were injected into the simulation at 449.5 km and recorded at 450.5 km, where the 100 km loss cone angle is approximately 67°. Left-side panels show number backscatter, and right-side panels show energy backscatter. Top panels show the pitch angle range of 0° to 90°, and bottom panels show a zoomed-in view on the pitch angle range from 60° to 70°.

put beam. The rates and sensitivity of backscatter presented in Figure 10 agree with previous work simulating backscatter (Marshall & Bortnik, 2018; Cotts et al., 2011).

The next parameter of interest is the pitch angle distribution (PAD) of backscattered electron populations. To visualize the dependence of backscattered PADs on pitch angle at a given energy, we calculate the backscattered PAD for each input pitch angle for which we have lookup tables available in units of electrons per steradian. This backscattered PAD is then normalized to integrate to unity over pitch angle for clarity of visualization, putting the PAD in units of electrons per steradian (PAD units) times degrees (unit from the normalization). This 1D distribution is then plotted horizontally on a heatmap with color representing the density of the distribution. This process was repeated for each input pitch angle at a given energy. The result of this procedure for two input energies is shown in Figure 11.

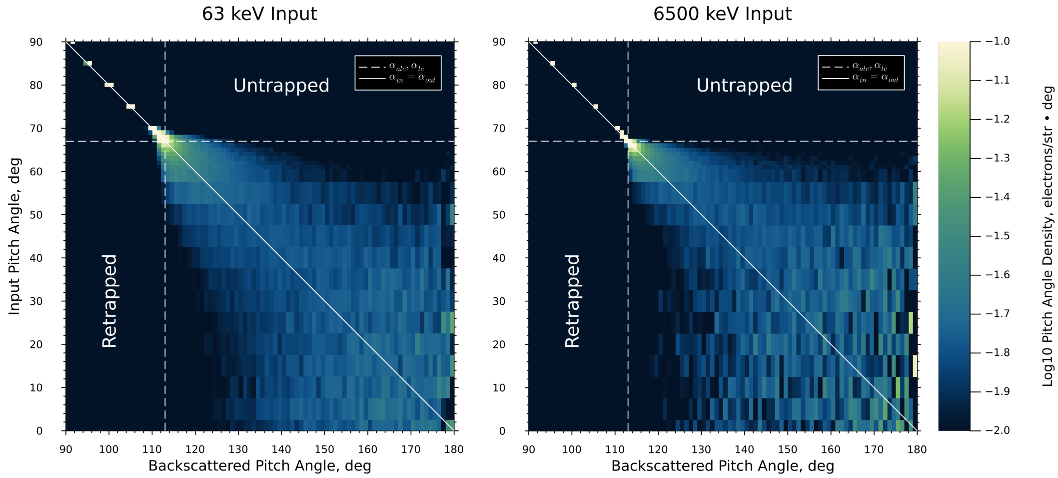

Figure 11. Input pitch angle vs. backscattered pitch angle derived from simulation at a variety of discrete input pitch angles. Each row represents the backscattered pitch angle distribution of a mono-pitch angle, monoenergetic input beam, normalized to solid angle coverage and to integrate to unity. Particles that started in the trapped region and were backscattered in the anti-loss cone are referred to as "untrapped", and particles that started in the loss cone and were backscattered in the trapped region are referred to as "retrapped". The 100 km loss cone angle in this simulation is approximately 67°, indicated by the dashed lines. The solid line indicates the location of adiabatic mirroring.

Figure 11 shows that for input pitch angles deep in the loss cone between approximately 0° and 50°, there is very little sensitivity or correlation between the backscattered PAD and the input pitch angle, with backscattered pitch angle distributions being roughly isotropic between 125° and 180°, and roughly symmetric around the magnetic conjugate of the input pitch angle (solid white line). In this range of input pitch angles the nominal magnetic mirror point is far below the surface of the Earth, and thus all backscatter is produced via collisions with atmospheric neutrals altering an electron's pitch angle sufficiently to escape the atmosphere, accounting for the isotropization of the backscattered distribution. For input angles between approximately 50° and 70°, we begin to see more sensitivity to input pitch angle with the backscattered PAD starting to skew toward the conjugate of the input pitch angle. This represents the regime where magnetic mirroring is beginning to influence the backscatter ratio and there are fewer atmospheric collisions occurring. Since the magnetic mirror point is increasing in altitude with increased input pitch angles (the mirror altitude goes above ground level at inputs around 60°), the pitch angle at any point in the atmosphere is closer to mirroring for these inputs than they were for the more field aligned inputs. Thus, it takes fewer collisions to adjust their trajectory to an upgoing direction, resulting in less pitch angle diffusion. This also correlates with the region of input pitch angles where backscattered energy begins to dramatically increase (see Figure 10), further implying a reduction in atmospheric collisions in this region. Within a few degrees of the 100 km mirror point at 67°, magnetic mirroring takes over completely, and these particles never encounter appreciable atmospheric density; thus, the backscattered pitch angle distribution is a nearly-perfect delta function about the magnetic conjugate of the input angle, lying directly on the $\alpha_{\text{in}} = \alpha_{\text{out}}$ line. We also note that the backscattered PAD does not depend much on energy, except in the approximately 60° to 65° region, where there is slightly more atmospheric scattering for the high energy beam than the low energy beam, potentially due to the deeper atmospheric penetration of high energy particles owing to their smaller angular deflections during collisions with neutrals.

Next, we observe the pattern of particle movement between regions. A particle that starts in the trapped region and is backscattered into the anti-loss cone is called "untrapped", and a particle starting in the loss cone and backscattering into the trapped region is called "retrapped". Figure 11 shows that there is negligible retrapping and untrapping in our simulation, with the vast majority of particles remaining in the region they started in (either the trapped region or the loss/anti-loss cone). There is a minor energy dependence to this observation, with the cutoff point varying by about 5° between the 63 keV input and the 6.5 MeV input. This energy dependence is consistent with previous work (Marshall & Bortnik, 2018). Moreover, we conclude that this predicts that using in-situ measurements of loss cone and anti-loss cone fluxes as a proxy measurement for precipitating and backscattered fluxes respectively is highly accurate.

The pitch angle distributions of backscattered electron beams has also been reported in Cotts et al. (2011); Marshall and Bortnik (2018), and Berland et al. (2023). The pitch angle distributions reported in Cotts et al. (2011) and Marshall and Bortnik (2018) (Figure 3 in both papers) show a very similar trend to our results, with field-aligned inputs having isotropized backscatter with a rapid transition to a delta function near the 100 km loss cone edge, with a slight energy dependence. Berland et al. (2023) predicts different pitch angle distributions (Figure 7 in their paper) that do not approach delta functions near the loss cone edge. This likely indicates a magnetic field configuration in their simulation that does not include magnetic mirroring.

Overall, we find that our simulation work agrees with previous modelling efforts and has been validated with in-situ data from the ELFIN mission.

8 Conclusions

Backscatter is an important part of the interconnected atmosphere-ionosphere-magnetosphere system. In this paper, we have used in-situ data from the ELFIN satellites to constrain EPP backscatter rates and have found that the backscatter ratio of precipitating electrons is primarily a function of the strength of loss cone filling. Backscatter rates were at or below 10% for the strongest filling of the loss cone ($J_{\mathrm{prec}}/J_{\mathrm{trap}} > 1$), increasing up to approximately 90% during periods with little to no loss cone filling. This variation is attributed to the pitch angle dependence of backscatter rates, with strong loss cone filling representing the backscatter rates deep in the loss cone and weak loss cone filling representing the backscatter rates near the edge of the loss cone.

We then introduced an updated EPP backscatter model which was used to simulate all ELFIN data that was used in the backscatter analysis. This model showed excellent prediction of backscatter due to precipitating fluxes in the trapped region and the near loss cone, with a tendency to overestimate backscattered at very low fluxes. This tendency was attributed to artifacts arising from ELFIN's noise floor and pitch angle resolution limiting the fidelity with which backscatter can be modelled.

Finally, we used our precipitation model to characterize the sensitivities and pitch angle distributions of backscatter. We found that backscatter is generally much more sensitive to input pitch angle than input energy, as previous work has shown. We found that backscatter is dominated by atmospheric interactions at more field aligned pitch angles, which isotropizes the backscattered pitch angle distribution in the anti-loss cone. In the region near the loss cone angle, magnetic mirroring begins to dominate over atmospheric interactions, leading to a backscattered pitch angle distribution that trends towards a delta function as the input pitch angle approaches the trapped region. We also found that particles that start in the trapped region are almost exclusively backscattered in the trapped region, and particles that start in the loss cone are almost exclusively backscattered in the anti-loss cone. These results suggest that very few electron-neutral interactions occur above 100 km. We further conclude that measured fluxes in the loss cone and anti-loss cone are good proxies for precipitating and backscattered populations respectively when measuring in-situ particle distributions.

8.1 Caveats and Future Work

The primary limitation of our model-data comparison is a lack of high-resolution data in sensitive regions of the pitch angle distribution, i.e. very near the loss cone, where resolution on the order of one degree is necessary to accurately characterize the precipitating and trapped populations. This limitation could possibly be avoided through the use of a fitting scheme. Such a scheme would utilize the recorded pitch angle and energy distribution and make an assumption for the driver of the precipitation. This driver could then be fed to a set of lookup tables containing archetypal pitch angle distributions for that driver. For example, Figure 3 in Capannolo et al. (2023) shows physics-based pitch angle distributions for EMIC-driven precipitation measured on ELFIN. These pitch angle curves could then be fitted to ELFIN's recorded flux and sampled at an arbitrarily high resolution. By predicting the shape of the pitch angle distribution in the sensitive region near the loss cone edge rather than assuming a stepped distribution based on the measured flux, we may recover a more accurate prediction of backscatter. This, however, introduces other uncertainties through the assumption of an underlying pitch angle distribution.

It is also possible to interpolate our backscatter lookup tables between nearby simulation runs to obtain backscattered particle distributions on a higher-resolution grid than the beams we calculated, as done in Cotts et al. (2011). Given the smooth variance of the backscatter parameters presented in this paper, interpolation would be a reasonable approach to produce a higher-fidelity backscatter model without excessive computational effort. In the trapped region, this interpolation would likely be as simple as assuming pure adiabatic mirroring at the conjugate pitch angle to the input angle, while a more complicated scheme would be required for the loss and anti-loss cone regions.

Open Research Section

ELFIN data is available for free at `https://data.elfin.ucla.edu/ela/`. Geant4 is available for free from `https://geant4.web.cern.ch/download/11.3.2.html`. The Geant4 code used to generate lookup tables is available for free at `https://github.com/julia-claxton/g4epp-source/tree/main`. The backscatter lookup tables generated by the Geant4 simulation are available for free at `https://zenodo.org/records/16542073`. All code used to perform the analysis and generate figures, as well as the list of ELFIN events used in the analysis, is available for free at `https://github.com/julia-claxton/2025_JGR_Backscatter`.

Acknowledgments

The authors acknowledge NASA award NNX14AN68G, NSF awards AGS-1242918 and AGS-2019950, and Vassilis Angelopoulos for use of ELFIN's EPD data. The authors thank the Ethan Tsai for providing technical support with ELFIN data, the Julia programming language team (Bezanson et al., 2017) for the use of the Julia language, the SpacePy Python package (Niehof et al., 2022; Steven K. Morley et al., 2010) for the use of their tools, and University of Colorado Research Computing for the use of their high-performance computing resources.

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